

Lab 01: Device Introduction and Configuration

Part 1: Device Introduction

SD Card and Card Reader:

A **MicroSD card** serves as the Raspberry Pi's primary storage device, containing the operating system, applications, configuration files, and user data required to run the system.

A **card reader** is used to connect the MicroSD card to a computer so the Raspberry Pi operating system can be installed or updated on the card.



Raspberry Pi 5

The **Raspberry Pi 5** is a powerful single-board computer featuring a 2.4 GHz quad-core 64-bit Arm Cortex-A76 processor, VideoCore VII GPU, and multiple RAM options. It supports dual 4K displays, USB 3.0, Gigabit Ethernet, Wi-Fi, Bluetooth, GPIO, and PCIe connectivity. Compared with the Raspberry Pi 4, it provides approximately **2-3 times higher CPU performance** and improved graphics and I/O capabilities. These improvements make it well suited for **IoT, robotics, automation, edge AI, and embedded-system applications**.



Power Adapter

A **45 W power adapter** provides stable and sufficient power to the Raspberry Pi and its connected USB devices and peripherals during operation.



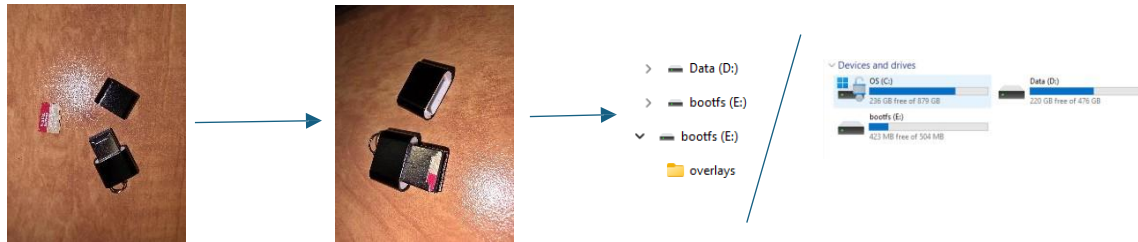
Mobile Hotspot

A **mobile hotspot** provides a shared Wi-Fi network that lets the Raspberry Pi and a computer connect to the same network, enabling remote access to the Raspberry Pi via tools such as **SSH or Remote Desktop**.

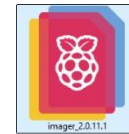


Part B: Setting Up Raspberry PI 5

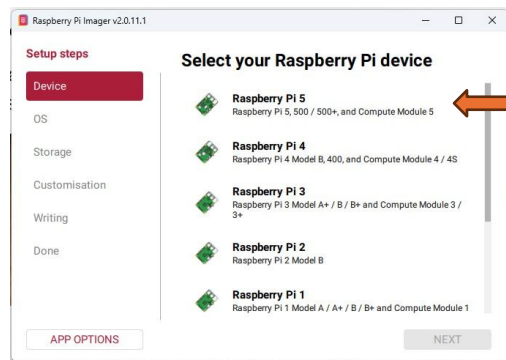
Step 1: Load the microSD card into the card reader, then insert the reader into a USB/USB-C port. Make sure the SD card is displayed in your computer.



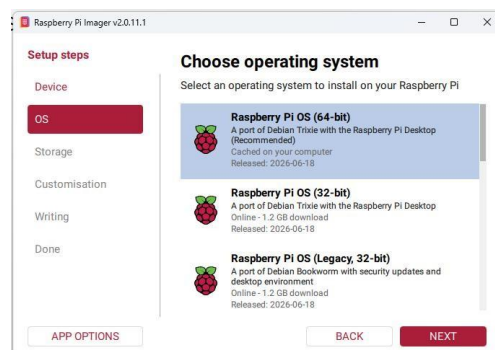
Step 2: Download and install [Raspberry Pi Imager](#) on your computer.



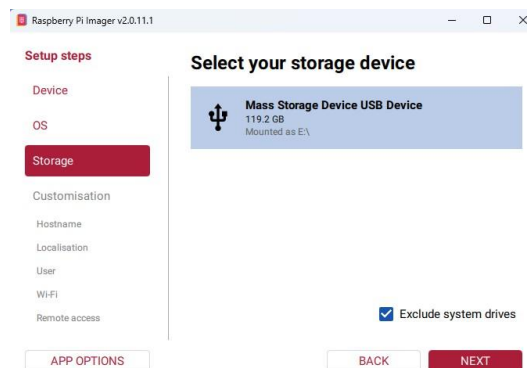
Step 3: Start the application and select Raspberry Pi 5 from the list of options.



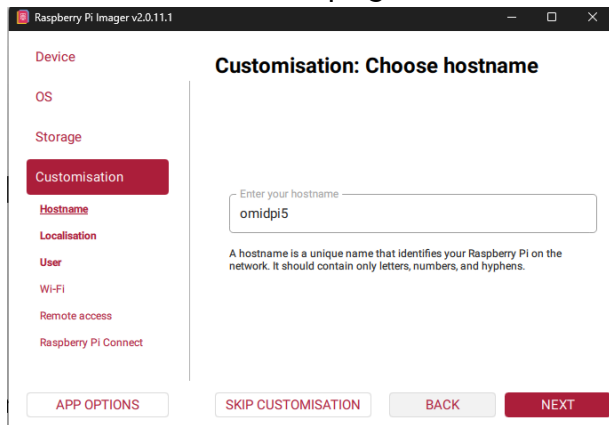
Step 4: Select Raspberry Pi OS (64-bit)



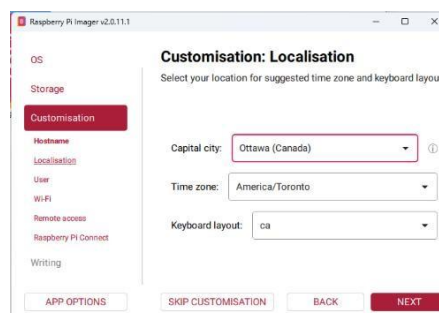
Step 5: Select the MicroSD Card Storage



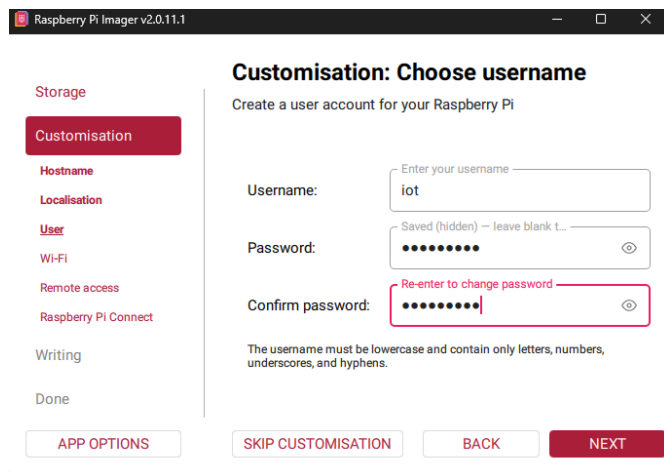
Step 6: Enter your Raspberry PI hostname. Please choose it carefully as later we will use this hostname to ssh/access/ping it from local computer.



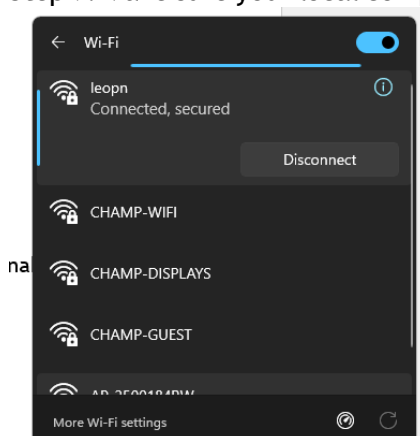
Step 7: Please entry the location information in this step.



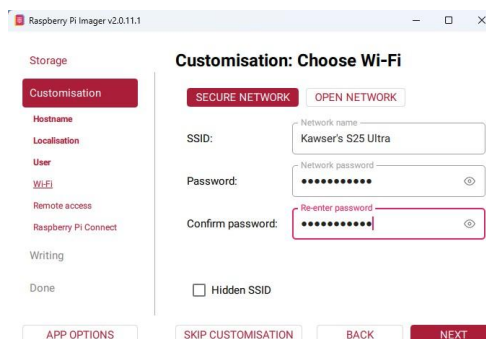
Step 8: Create username/account for the Raspberry pi OS. Without it we cannot log in to the Raspberry PI.



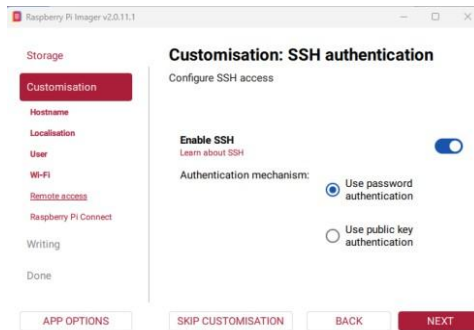
Step 9: Make sure your local computer is connected to the Mobile Hotspot.



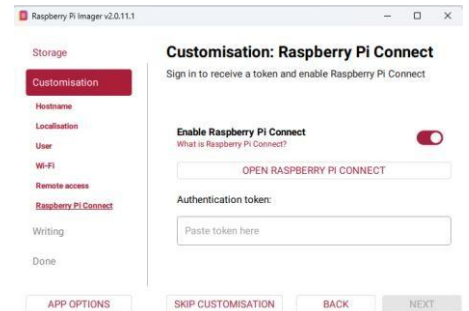
Step 10: Add your Mobile Hotspot Wi-Fi information in the PI imager Wi-Fi section.



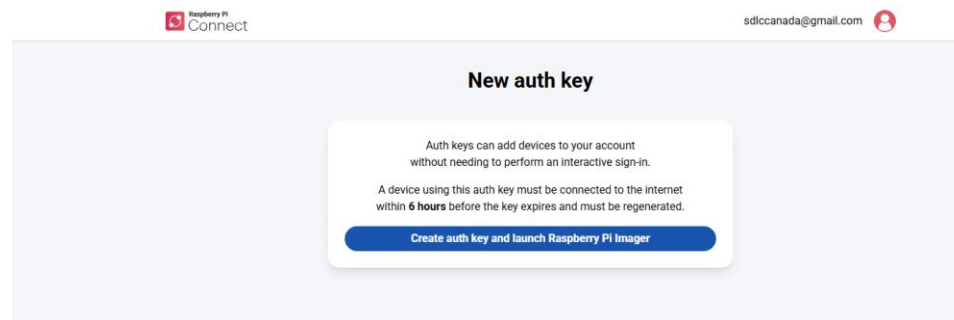
Step 11: Enable SSH to access Raspberry PI remotely (from your laptop or other SSH enabled devices)



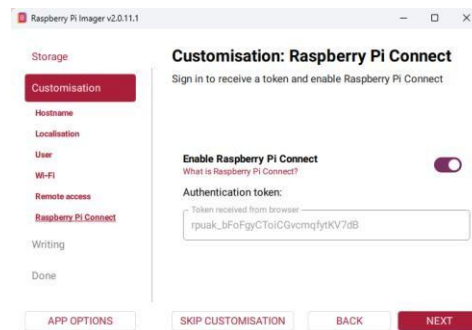
Step 12: Enable Pi Connect to access the device from anywhere, click on the Open Raspberry Pi Connect. This step can **be skipped** if you don't keep your PI powered on all time.



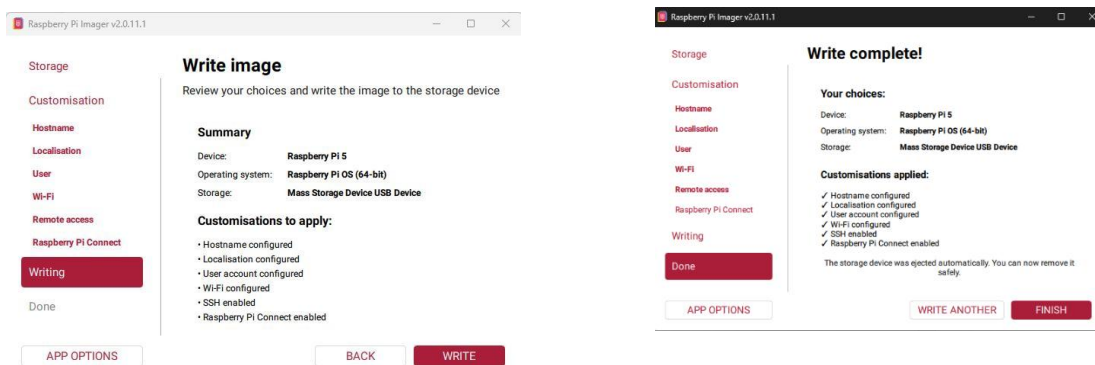
Step 13: Create an authentication key to register your PI. For the first time it will automatically create the authentication key for you. (skip this if you skip step 12).



Step 14: Once authenticated, you will see the authentication token generated in the PI Imager. Click Next. (Skip if you skipped step 12 C 13)



Step 15: Click on Write Image. It will erase the MicroSD card and Write your configured OS in it. Wait patiently until your image write is done.



-----> DONE

Part C: Connecting to Raspberry PI

Step 1: Once OS setup is finished, please take out the card reader from computer USB port, take the MicroSD card out and put it in the Raspberry PI MicroSD card port. Power On your Raspberry PI now. Make sure your Mobile Phone hotspot is still active. Raspberry Pi will automatically get connected to your mobile hotspot.

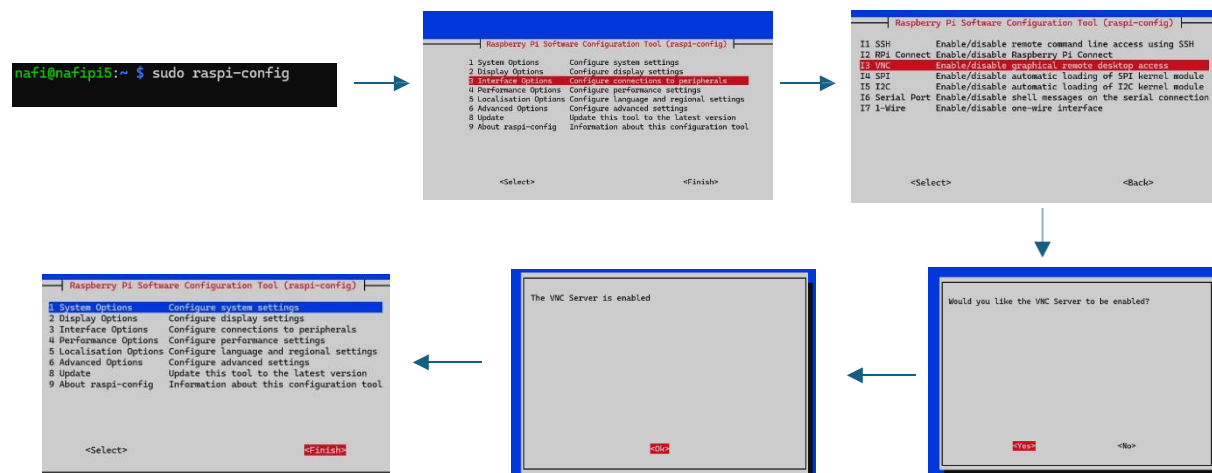
Step 2: Open your Local Machine/ Laptop Terminal and ssh with your set username and raspberry pi name in the following way:

```
ssh username@hostname
```

```
PS C:\Users\opanahi> ssh iot@teacher-05
The authenticity of host 'teacher-05 (fe80::8aa2:9eff:fe25:4f09%18)' can't be established.
ED25519 key fingerprint is SHA256:uthwzaWkOU7kuqoEMZzvp1pxo2vMx2xascOCJVWaQ5A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Now we need to enable VNC and setup VNC server so that we can use the GUI of Raspberry PI from the Local Machine (laptop).

Step 3: Use the `sudo raspi-config` command and follow the steps shown in the following images: (use keyboards arrows and tab to switch between options, enter to select)



Step 4: use the following two commands to prepare the raspberry-pi environment

```
sudo apt update
sudo apt upgrade (put Y if prompt to install)
```

Step 5: to access raspberry using remote desktop provided in Windows OS, use the following commands one after another to install xrdp server. Without xrdp, windows can not access raspberry pi OS. If you want to access other VNC client, like readVNC or tigerVNC, you can skip everything from this step until VNC connection establishment.

```

sudo apt update
sudo apt install --reinstall xrdp xorgxrdp (put Y when prompted)
sudo adduser xrdp ssl-cert
sudo systemctl restart xrdp
sudo systemctl status xrdp
  
```

The last command should show the following output:

```

iot@teacher-05:~$ sudo adduser xrdp ssl-cert
iot@teacher-05:~$ sudo systemctl restart xrdp
iot@teacher-05:~$ sudo systemctl status xrdp
● xrdp.service - xrdp daemon
   Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-02 11:36:02 EDT; 6s ago
     Invocation: 9646e5c8be9040baa65800770bcad930
       Docs: man:xrdp(8)
            man:xrdp.ini(5)
    Main PID: 362723 (xrdp)
      Tasks: 1 (limit: 19413)
        CPU: 21ms
       CGroup: /system.slice/xrdp.service
              └─362723 /usr/sbin/xrdp --nodaemon

Sep 02 11:36:02 teacher-05 systemd[1]: Starting xrdp.service - xrdp daemon...
Sep 02 11:36:02 teacher-05 systemd[1]: Started xrdp.service - xrdp daemon.
Sep 02 11:36:02 teacher-05 xrdp[362723]: [INFO ] starting xrdp with pid 362723
Sep 02 11:36:02 teacher-05 xrdp[362723]: [INFO ] address [0.0.0.0] port [3389] mode 1
Sep 02 11:36:02 teacher-05 xrdp[362723]: [INFO ] listening to port 3389 on 0.0.0.0
Sep 02 11:36:02 teacher-05 xrdp[362723]: [INFO ] xrdp_listen_pp done
iot@teacher-05:~$ |

```

Step 6: You have to create a new sudo user. For some unknown reason, Xrdp doesn't work with default user login and password after step 5. Use the following commands to create your user.

```

sudo adduser new_user (enter new password and other information,
for simplicity you can choose admin as user and iotadmin as
password) sudo usermod -aG sudo new_user
groups new user

```

```

iot@teacher-05:~$ sudo adduser admin
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for admin
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] |

```

```

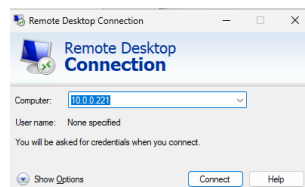
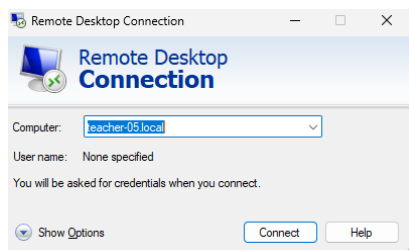
iot@teacher-05:~$ sudo usermod -s /dev/null admin
iot@teacher-05:~$ groups admin
admin : admin sudo users
iot@teacher-05:~$ ifconfig
eth0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 88:a2:9e:25:4f:08 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 106

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    Loop txqueuelen 1000 (Local Loopback)
    RX packets 17 bytes 2704 (2.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 17 bytes 2704 (2.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 19.0.0.221 netmask 255.255.255.0 broadcast 19.0.0.255
    inet6 fe80::8a2:9eff:fe25:4f09 prefixlen 64 scopeid 0x20<link>
    ether 88:a2:9e:25:4f:09 txqueuelen 1000 (Ethernet)
    RX packets 533968 bytes 869779768 (872.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 79821 bytes 6060762 (5.7 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

Step 7: Start your VNC client or Remote Desktop Connection (Windows User) application. For Computer, user your raspberry host_name.local or raspberry pi assigned IP address. To know the raspberry pi IP address, user ifconfig command



Click on connect, accept the ssh certificate and ready to user Xrdp VNC server.

